

4 hours on examples

Problem 3 (KSATST)

We use induction to prove that, for any n , there exists positive integers $a_0, a_1, \dots, a_n \geq 2$ such that,

$$a_0! = a_1! a_2! \dots a_n!$$

With the addition requirement that the LHS is greater than 3.

Base Case: $n = 1$

We let $a_0 = a_1 = 3$ to get $3! = 3!$.

Inductive Step: $n = k$

With a construction for,

$$a_0! = a_1! a_2! \dots a_{k-1}!$$

We let $a_k = a_1! a_2! \dots a_{k-1}! - 1$, which must be greater than 2.

$$a_1! a_2! \dots a_{k-1}! a_k! = (a_1! a_2! \dots a_{k-1}!) \cdot (a_1! a_2! \dots a_{k-1}! - 1)! = (a_1! a_2! \dots a_{k-1}!)!$$

The result is still a factorial, so there exists a new $a_0 = a_1! a_2! \dots a_{k-1}!$ that satisfies this.

Conclusion: By mathematical induction, the claim is proved for all positive n , including $n = 100$ as desired.

Problem 4 (10JMO4)

The restriction I used was $(0, 0)$.

Base cases $n = 0, 1, 2$:

For $n = 0$, notice the triangle with vertices $(0, 0); (-1, 1); (1, 1)$ has area $(2^0 \cdot 1)^2$.

For $n = 1$, notice the triangle with vertices $(0, 0); (-1, 1); (8, 64)$ has area $(2^1 \cdot 3)^2$.

For $n = 2$, notice the triangle with vertices $(0, 0); (24, 576); (6, 36)$ has area $(2^2 \cdot 9)^2$.

Inductive step: $n = k \rightarrow n = k + 3$

For a given triangle with area $(2^k m)^2$ satisfying the conditions with coordinates $(x_1, x_1^2); (x_2, x_2^2); (x_3, x_3^2)$, notice if we quadruple x_1, x_2, x_3 , the area is 2^6 times what it was, so we have a new triangle with area $(2^{k+3} m)^2$ still with integer coordinates.

Conclusion: By mathematical induction there exists such a triangle for all n .

Problem 6 (23MPO1)

We start with $(p_1, p_2, \dots, p_n) = (1, 2, \dots, n)$.

For this choice of $p_1, \dots, p_n$, we define $0 \leq r < n$ as the remainder when one divides $p_1 + 2p_2 + \dots + np_n$ by n . Note that if we find a sequence of swaps on $(p_1, \dots, p_n)$ which decreases the sum by r , we are done.

Note that when we swap p_i and p_{i+1} , the original sum was,

$$\sum_{k=1}^{i-1} k^2 + i^2 + (i+1)^2 + \sum_{k=i+2}^n k^2$$

While the new sum is,

$$\sum_{k=1}^{i-1} k^2 + i(i+1) + (i+1)i + \sum_{k=i+2}^n k^2$$

The new sum is hence 1 less as $2i(i+1) - (i^2 + (i+1)^2) = -1$ no matter the choice of i .

Similarly, when we swap p_i and p_{i+2} , the new sum is 4 less than the original no matter the choice of i .

Below is the construction to decrease the sum by r written in pseudo-code. (Is pseudo-code perfectly acceptable always or should I save it as a last resort?)

```
int lmove :=  $\left\lfloor \frac{r}{4} \right\rfloor$ ;
int smove := the remainder when one divides  $r$  by 4 (i.e.  $r \% 4$ );
int marker = 1;
Repeat (lmove) times {
    swap( $p_{\text{marker}}$ ,  $p_{\text{marker}+2}$ ); // Decreases the sum by 4.
    Increase marker by 3;
}
Repeat (smove) times {
    swap( $p_{\text{marker}}$ ,  $p_{\text{marker}+1}$ ); // Decreases the sum by 1.
    Increase marker by 2;
}
```

We can check that we will never go out of bounds. We will perform the lmoves at most $\left\lfloor \frac{r}{4} \right\rfloor < \frac{n}{4}$ times with each time increasing the marker by 3. Therefore, after all the lmoves are done, we've only gotten through $\frac{3n}{4}$ of the n elements, leaving $\frac{n}{4} > 6$ elements untouched. We then perform the smoves at most 3 times with each using 2 elements at most for 6 in total, so we cannot run out.

Since we never swap the same number or position twice, after we do this operation, the desired sum will be a multiple of n .

7 club; 6 hours

Problem 8 (13BRA2)

This is yet another new topic so can you provide a score or feedback for this proof? Thanks!

It is sufficient to prove that, for any integer n , we may pick a positive integer b such that the number of divisors of ab over all divisors a of n is distinct. This means that we may uniquely determine a . We may then take n to be the *lcm* of all the elements of A to be done.

We prove that there exists a b using induction on the number of prime divisors of n .

Base case: $m = 1$

Selecting $b = 1$ works as all the divisors of n will still be powers of the original prime. Each will have a distinct number of divisors.

Inductive step: $n = k - 1$ to $n = k$

We let the prime factorization of n be,

$$n = \sum_{i=1}^k p_i^{e_i}$$

We let b' be the solution for the first $k - 1$ prime divisors of n , or the number that satisfies the induction hypothesis for,

$$n' = \sum_{i=1}^{k-1} p_i^{e_i}$$

Let $S' = \{s_1, s_2, \dots\}$ be the set of the unique number of divisors of $a'b'$ over the divisors a' of n' . We claim that the following b works for a sufficiently large M ,

$$b = b' \cdot p_k^M$$

The new set S of divisors of ab over the divisors a of n is hence,

$$\begin{aligned} S &= \bigcup_{i=M+1}^{M+e_k+1} \{s_1 \cdot i, s_2 \cdot i, \dots, s_n \cdot i\} \\ &= \bigcup_{s_i \in S'} \{(M+1)s_i, (M+2)s_i, \dots, (M+e_k+1)s_i\} \end{aligned}$$

By Pigeonhole, if the size of the new set is $(e_k + 1) \cdot |S'|$, then everything would be unique, and we are done. WLOG, sort $s_1 < s_2 < \dots < s_n$. Note that,

$$\begin{aligned}
(M+1)s_{i+1} &> (M+e_k+1)s_i \\
\iff (M+1)(s_i+1) &> (M+e_k+1)s_i \\
\iff M+1 &> e_k s_i
\end{aligned}$$

Therefore, we may pick a $M \geq e_k \max(S')$ to be done.

Conclusion: By mathematical induction it is possible to choose such a b .

Problem 9 (22MPO2)

Note that scaling the trapezoid does not affect the result. We let,

$$A = (0, 0); C = (1, 0)$$

We add the restriction that $AB \perp AC$ and $CD \perp AC$ (losing 1 degrees of freedom, which is ok), so let $B = (0, x)$ and $D = (1, y)$. Clearly having D on the same side of AC as B does not work. Therefore, let D be on the other side of AC . Add yet another restriction that $AB + CD = AC$, so $y = x - 1$. Therefore, $CD = 1 - x$ and $AB = x$.

Looking at the diagram, $AB + BC = BD$ seems promising. Therefore,

$$\begin{aligned}
BD &= \sqrt{1+1} = x + \sqrt{1+x^2} \\
\iff \sqrt{2} - x &= \sqrt{1+x^2} \\
\iff x^2 - 2x\sqrt{2} + 2 &= x^2 + 1 \\
\iff x &= \frac{\sqrt{2}}{4}
\end{aligned}$$

We claim that it is possible. We employ cartesian coordinates for the construction. The quadrilateral is,

$$A = (0,0); B = \left(0, \frac{\sqrt{2}}{4}\right); C = (1, 0); D = \left(1, \frac{\sqrt{2}}{4} - 1\right)$$

We see that,

$$\begin{aligned}
AB + CD &= 1 = AC; \\
AB + AC &= \frac{\sqrt{2}}{4} + \sqrt{1 + \frac{1}{8}} = \frac{\sqrt{2}}{4} + \frac{3\sqrt{2}}{4} = \sqrt{2} = \sqrt{AC^2 + (AB + CD)^2} = BD
\end{aligned}$$

15 clubs; 8 hours

Problem 10 (17MPO2)

Case 1: n is odd.

We let $P(x) = \frac{x^n - 1}{x - 1} = x^{n-1} + x^{n-2} + \dots + 1$ and $Q(x) = 0$. Note that P is non-negative because $P(1) = 1$ and $x^n - 1, x - 1$ have the same signs for $x \neq 1$.

Case 2: n is even.

Both sides are divisible by $(1 - x)$, so $1 - x \mid Q(x)$. However, since Q is non-negative, we must also have $(1 - x)^2 \mid Q(x)$. Similarly, $1 + x$ divides both sides, so $(1 + x)^2 \mid P(x)$.

Let $q(x) := \frac{Q(x)}{(1-x)^2}; p(x) := \frac{P(x)}{(1+x)^2}$.

$$1 - x^n = (1 - x)(1 + x)^2 p(x) + (1 - x)^2 (1 + x) q(x)$$

$$\Leftrightarrow x^{n-2} + x^{n-4} + \dots + 1 = (1 + x)p(x) + (1 - x)q(x)$$

We may sufficiently and necessarily guarantee p, q to be non-negative. Clearly, picking,

$$p(x) = q(x) = \frac{x^{n-2} + x^{n-4} + \dots + 1}{2}$$

We are done.

We pick,

$$Q(x) = (1 - x)^2 \left(\frac{x^{n-2} + x^{n-4} + \dots + 1}{2} \right);$$

$$P(x) = (1 + x)^2 \left(\frac{x^{n-2} + x^{n-4} + \dots + 1}{2} \right);$$

Therefore,

$$1 - x^n = (1 - x)P(x) + (1 + x)Q(x)$$

$$\Leftrightarrow x^{n-2} + x^{n-4} + \dots + 1 = (1 - x) \left(\frac{x^{n-2} + x^{n-4} + \dots + 1}{2} \right) + (1 + x) \left(\frac{x^{n-2} + x^{n-4} + \dots + 1}{2} \right)$$

This is clearly true. Now note that, since $(1 - x)^2$ is positive and $\frac{x^{n-2} + x^{n-4} + \dots + 1}{2}$ is positive as it is a sum of even-degree terms with positive coefficient, we have Q, P are both positive.

Problem 12 (15RMM1)

We claim it is possible.

By symmetry, it is sufficient to prove $\gcd(a_n, a_{n+1}) = 1$ for all $n \geq 1$ and $\gcd(a_n, a_m) > 1$ for all $n > m + 1$. The construction is as follows, we start with the sequence,

$$2 \cdot 3, 5 \cdot 7, 11 \cdot 13, 17 \cdot 19, 23 \cdot 29, 31 \cdot 37, \dots$$

a_n is the product of the $(2n - 1)$ th and $(2n)$ th primes. We have that $\gcd(a_n, a_{n+1}) = 1 \forall n \geq 1$. For the next step, over all $a_n = p_{2n-1} \cdot p_{2n}$, we multiply $a_{n+2}, a_{n+4}, \dots$ by p_{2n-1} and $a_{n+3}, a_{n+5}, \dots$ by p_{2n} .

To demonstrate,

$$a_1 = (2 \cdot 3);$$

$$a_2 = (5 \cdot 7);$$

$$a_3 = 2 \cdot (11 \cdot 13);$$

$$a_4 = 3 \cdot 5 \cdot (17 \cdot 19);$$

$$a_5 = 2 \cdot 7 \cdot 11 \cdot (23 \cdot 29);$$

$$a_6 = 3 \cdot 5 \cdot 13 \cdot 17 \cdot (31 \cdot 37);$$

...

Finish proof with check that both conditions are still met.

21 clubs; 10 hours

Problem 14 (09SLC3)

Let $A_i = 2^i a_i; B_i = 2^i b_i$. We have,

$$A_{i+1} = \begin{cases} 8A_{i-1} + 6A_i & \text{if } \epsilon_i = 0 \\ 12A_{i-1} + 2A_i & \text{if } \epsilon_i = 1 \end{cases}; A_0 = 1; A_1 = 14$$

We have $a_n = b_n \Leftrightarrow A_n = B_n$, so we prove that instead.

We employ double counting. Imagine we have a long building with n rooms in a row, each of the $n - 1$ pairs of adjacent rooms separated by a wall. Each room is colored one of 14 colors labeled $1, 2, \dots, 14$. We force the first room to be labeled one.

For each wall, we label it a 0-wall or a 1-wall depending on whether there is a door installed on it.

For two rooms separated by a 0-wall, at least one of the following conditions must be met:

- At least one of the rooms are colored 1.
- The two room's colors are $i, j \neq 1$, then $i - j \equiv -2, -1, 0, 1, 2 \pmod{13}$.

For two rooms separated by a 1-wall, at least one of the following conditions must be met:

- At least one of the rooms are colored 1.
- The two rooms are colored the same.

We employ dynamic programming to calculate the possible colorings. Let ϵ_i denote the type of wall between rooms i and $i + 1$. For $\epsilon_i = 0$, if the right room is colored 1, or not colored 1 but is $-2, -1, 0, 1, 2 \pmod{13}$ from the previous, there are a total of $6A_i$ options. If not, then the left room must be colored 1, granting $8A_{i-1}$ options. We do similar operations on $\epsilon_i = 1$ to get,

$$A_{i+1} = \begin{cases} 8A_{i-1} + 6A_i & \text{if } \epsilon_i = 0 \\ 12A_{i-1} + 2A_i & \text{if } \epsilon_i = 1 \end{cases}; A_0 = 1; A_1 = 14$$

By symmetry, starting from the left to the right gives the same number of results. If b_i is how many ways there are to color from the right i most rooms where the rightmost room needs to be colored 1 instead.

$$B_{i+1} = \begin{cases} 8B_{i-1} + 6B_i & \text{if } \epsilon_{n-i} = 0 \\ 12B_{i-1} + 2B_i & \text{if } \epsilon_{n-i} = 1 \end{cases}; B_0 = 1; B_1 = 14$$

Clearly, these two are counting symmetric things, so $A_n = B_n$ and we are done.

Problem 15 (14TWNQ3J1)

We claim that this is impossible.

Notice that every row, column, and diagonal sum to $\frac{1}{6} \left(\frac{36 \cdot 37}{2} \right) = 111$. Hence, it is sufficient to prove that it is impossible to fill a grid with positive integers such that every row, column, and diagonal sum to an odd number.

This is equivalent to filling a 6×6 grid with binary numbers representing the parity of each cell. There must be an odd number of 1's in each row, column, and diagonal.

A	B	A	B	A	B
	C		C		C
A	B	A	B	A	B
	C		C		C
A	B	A	B	A	B
	C		C		C

Let a, b, c be the sum of the numbers at positions marked by A, B, C respectively,

$$a + b \equiv 1 \pmod{2};$$

$$b + c \equiv 1 \pmod{2};$$

$$c + a \equiv 1 \pmod{2}$$

Adding these equations, the LHS is even while the RHS is odd, so impossible.

29 clubs; 15 hours

Problem 17 (99AMO5)

Note that a player may win in one move if and only if, the board has a $[S][O][]$, $[S][][S]$, or $[][O][S]$ since they must complete the $[S][O][S]$. We call these three arrangements “losing”.

Also, whenever we have $[S][][][S]$, which we call a “zugzwang”, whenever a player writes either letter in either of the empty boxes between the S ’s, they’d make a “losing” arrangement.

Whenever a player creates a “losing” arrangement or plays in a “zugzwang”, we call it a “blunder” (even if they were forced to) do it.

The construction is as follows:

Case 1: Player 1 places an S .

We place another S three cells away to create a “zugzwang”. This is always possible in at least one direction.

Case 2: Player 1 places an O .

We locate a space of seven empty cells and place an S in the center. No matter what player one’s next move is, we place another S three cells away to create a “zugzwang”. This is always possible in at least one direction.

In either case, we proceed with the game by either,

- Play an O adjacent to another O .
- Play an O adjacent to nothing.
- Play an S adjacent to another S in a position outside of a “zugzwang”.

Note that none of these three moves will “blunder”.

We claim that this is always possible if the first player does not “blunder” and will eventually cause player one to run out of non-“blunder” moves on their turn.

Consider the time when we are unable to play any of these three moves. Note that this will eventually happen as the board will be filled further and further.

Whenever we have a gap of three empty spaces or more, we may place an O in the center. Whenever we have a gap of two empty spaces, there must be an S on either side or else we can place an O adjacent to another O . Whenever we have a gap of one space, there similarly must be an S on both sides, but then we may simply place an S in the middle.

Therefore, in such a situation, the board will be filled except for two-space gaps that are “zugzwangs”. However, an even number of empty squares means an even number of moves played. Therefore, when we eventually run out of moves it will be player one’s turn. Furthermore, since we’ve constructed at least one “zugzwang”, the game is not over yet and player one will now be forced to play in a “zugzwang” and hence lose the game.

Problem 18 (16IMO5)

We claim the answer is 2016.

Proof minimum:

We must erase at least one of each of $(x - 1), (x - 2), \dots, (x - 2016)$ for if not, we’d have a certain $(x - a)$ in both sides. Plugging in $x = a$ we have an equality. Therefore, we need to erase at least 2016.

Proof attainability:

On the LHS, we erase factors $(x - a)$ for all $a \equiv 0, 1 \pmod{4}$. On the RHS, we erase all the $2, 3 \pmod{4}$.

$$(x - 1)(x - 4)(x - 5)(x - 8) \dots (x - 2016) = (x - 2)(x - 3)(x - 6) \dots (x - 2015)$$

It is sufficient to show that,

$$(x-1)(x-4)(x-5)(x-8) \dots (x-2016) < (x-2)(x-3)(x-6) \dots (x-2015)$$

If x is an integer, either the RHS is zero and LHS negative or LHS negative and RHS zero. Assuming that x is not an integer.

Note that,

$$(x-1)(x-4) < (x-2)(x-3);$$

$$(x-5)(x-8) < (x-6)(x-7);$$

...

$$(x-2013)(x-2016) < (x-2014)(x-2015)$$

If $[x] \equiv 0 \pmod{4}$, then everything is positive so multiply and we're done.

If $[x] \equiv 1, 3 \pmod{4}$ then exactly one expression on the LHS will be negative and everything else is positive so we're also done.

Finally, consider the case where $[x] \equiv 2 \pmod{4}$. There is one equation with negatives on both sides, say,

$$(x-a-3)(x-a) < (x-a-1)(x-a-2)$$

After multiplying, both sides will be negative, so we must instead prove,

$$|(x-1)(x-4)(x-5)(x-8) \dots (x-2016)| > |(x-2)(x-3)(x-6) \dots (x-2015)|$$

For some integer m , we let $x \in (4m-2, 4m-1)$.

On this range, the ratio $\left| \frac{(x-4m-3)(x-4m)}{(x-4m-1)(x-4m-2)} \right| \geq 9$.

Claim: $\prod_{k \geq 0} \frac{(4k+2)(4k+3)}{(4k+1)(4k+4)} < e$

We have,

$$\begin{aligned} \prod_{k \geq 0} \frac{(4k+2)(4k+3)}{(4k+1)(4k+4)} &< e \\ \Leftrightarrow \prod_{k \geq 0} \frac{16k^2 + 20k + 6}{16k^2 + 20k + 4} &< e \end{aligned}$$

Taking log of both sides,

$$\Leftrightarrow \sum_{k \geq 0} \ln \left(1 + \frac{2}{(4k+1)(4k+4)} \right) < 1$$

Since $\ln(1+x) < x$ for all positive x ,

$$\Leftrightarrow \sum_{k \geq 0} \frac{2}{(4k+1)(4k+4)} < 1$$

$$\Leftrightarrow \frac{1}{2} + \sum_{k \geq 1} \frac{1}{8k^2} < 1$$

$$\Leftrightarrow \frac{1}{2} + \frac{\pi^2}{48} < 1$$

This is clearly true.

Using the claim, we'd get, $\prod_{k=i}^n \frac{(4k+2)(4k+3)}{(4k+1)(4k+4)} < e$, where k is still incremented by one, for any non-negative real number i, n . Applying this,

$$\begin{aligned} & \left| \frac{(x-1)(x-4)(x-5)(x-8) \dots (x-2016)}{(x-2)(x-3)(x-6) \dots (x-2015)} \right| > \frac{9}{e^2} \\ \Rightarrow & \left| \frac{(x-1)(x-4)(x-5)(x-8) \dots (x-2016)}{(x-2)(x-3)(x-6) \dots (x-2015)} \right| > 1 \\ \Leftrightarrow & |(x-1)(x-4)(x-5)(x-8) \dots (x-2016)| > |(x-2)(x-3)(x-6) \dots (x-2015)| \end{aligned}$$

As desired.

37 clubs; 17 hours

Problem 19 (22IGO4)

We show a construction building the polygons P, Q using k unit squares each.

We start with the m -sided polygon P , the one with more sides. We start by building a "base" which is a $1 \times m$ rectangle of m adjacent unit squares connected to each other. This has 4 sides.

We next attach $\frac{m-4}{2}$ more unit squares connected to the "base" on the long side, with each of the unit separated from each other and not touching any of the corners of the base. Each of these additions will hence add two more sides to the polygon and we'll have m in the end.

To construct our polygon with n sides Q , we first duplicate P and move $\frac{m-n}{2}$ of the unit squares sticking out from the "base" to the end of the "base" to extend it. Each of these operations will remove two sides from the polygon, and we'll have n in the end.

Problem 21 (CF1917E)

We claim that there is no solution if and only if (k is odd) OR ($(N \neq 2)$ AND ($k = 2$ OR $n^2 - 2$)).

Sketch: Odds don't work clearly, (N, k) work $\Leftrightarrow (N, N^2 - k)$ work by just flipping every bit. We can quickly check $k = 2$ clearly doesn't work if $N \neq 2$.

For construction, if $k > \frac{N^2}{2}$ we first flip all the bits such that $k \rightarrow N^2 - k$. Then, if $k \equiv 2 \pmod{4}$, we add a ,

1 1 0

1 0 1

0 1 1

Square on the corner. We then just start filling in the rest of the grid with,

1 1

1 1

We can check that there's enough space since $2k \leq N^2$ and we're done. All the rows/columns have even sums.

43 clubs; 18.5 hours

Problem 23 (10SLC1)

We claim that it is possible. The construction is as follows:

Singer 1, 2, 3 must respectively perform later than signers 4 ... 20. Isolating 1, 2, 3, there are $3!$ ways of arranging them independently.

(Played around with this for too long and couldn't find one that worked :C)

For singers 4, 5, ..., 14, we have,

- Singer 5 after 4, 6 after 5, ..., 9 after 8
- Singer 11 after 10, 12 after 11, 13 after 12, and 14 after 13
- Singer 14 after 4
- Singer 5 after 10

The first two conditions give $\binom{11}{5}$, third condition removes 1, and last condition removes $\binom{9}{5}$. In total, we have 335 ways.

Note that, as a group, singers 1, 2, 3 must go after the group 4, 5, 6, 7, 8, who then must go after the group 9, ..., 20. We can hence consider them disjointly and independently, multiplying together how many ways those groups may rearrange themselves. In total, that's $6 \cdot 335 = 2010$ ways.

Problem 24 (13RMM2)

We claim that choosing $g(x) = x^2$; $h(x) = x + 1$ works.

We hence have,

$$f(x^2) = f(x)^2; f(x + 1) = f(x) + 1$$

First note that obviously $f(x) \geq 0 \forall x \geq 0$ as $f(x)^2 \geq 0$.

Claim: $x - f(x) < 1 \forall x > 0$

If not, we can shift x until it is on $[0, 1)$, by which time $f(x) < 0$, so contradiction.

Claim: $f(x) \geq x$

For any positive x such that $f(x) < x$, we repeatedly use $f(x^2) = f(x)^2$ until we get to $f(x^{2^n}) < x^{2^n} - 1$, getting a contradiction. Therefore, $f(x) \geq x$ for all positive x . Then, we may shift the negatives as well to get this for all real x .

Claim: $f(x) \leq 0$ for $x \leq 0$

If not, shift until we have $-1 < f(x - c) \leq 0$, then square it to arrive at a contradiction.

Now, since $f(-x)^2 = f(x)^2$ but the negative is ≤ 0 , we have $f(x) = -x$. Since $f(x) \geq x$ for all real x , though, $f(x) \equiv x$.

57 clubs; 21.5 hours